Smart ICT Tool overview

Revision 1.0.0 from 10.02.2021

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1 Concept

The purpose of this tool is to determine the testcoverage of digital circuit diagrams which are being provided as the ODB++ format. Actually there is only the possibility to determine boundary scan related test coverage results. Furthermore it is possible to store and reaload as well imported data from the ODB++ files as settings files for later usage.

2 Components

The software itself is separated into two logic independent tools which need to be started in order to make working with it possible.

- Smart ICT GUI - The graphical representation of the tool which provides the possibility of triggering different kind of actions to the user by mouse and keyboard
- The GRPC server for getting all necessary components by using the PCB investigator - This is a GRPC server which could be started separately on each possible machine and uses the API of the PCB investigator for collecting all necessary objects out of the ODB++ project. This program has to be started on a PC where a valid license for using the PCB investigator is present.

In order for using the full power of the Smart ICT tool it is highly recommended to start and connect both tools together.

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3 Overview

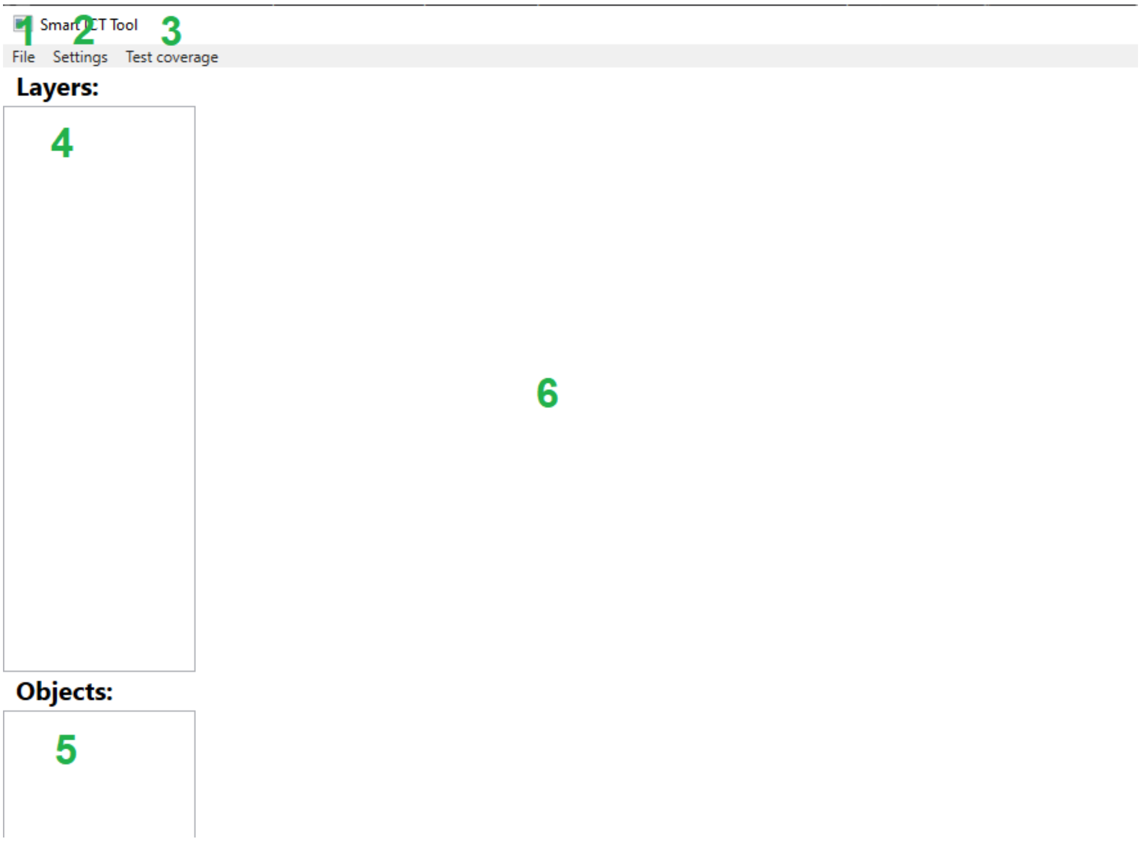

Figure 1: The overview of the application

The graphical Smart ICT tool itself consists of 6 different areas for different actions or representations

1. File menu item - Within this menu item you have the possibility to trigger actions related to ODB++ file data
2. Settings menu item - Within this menu item you have an overview of all available settings of the tool
3. Test coverage menu item - Within this menu item you have all test coverage related actions
4. Layers view - An overview of all available layers of the ODB++ project
5. Objects view - An overview of all available test related object groups
6. Painting area - The area where all electrical objects of the ODB++ project will be painted at

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3.1 File menu item

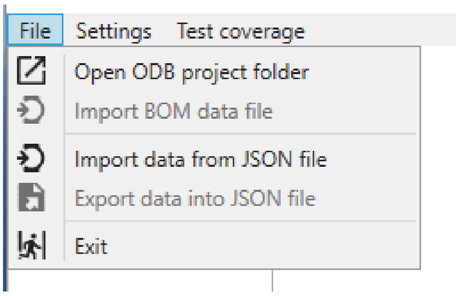

Figure 2: The file children menu items

Within the file menu item you will find the following items in order from top to bottom

- Open ODB project folder - By pressing this item a new folder selection dialog will appear to make the user select any folder containing valid ODB project files
- Import BOM data file - This item will only be available after you have loaded a valid project into the tool and provides the possibiliy of reading out any BOM related files in order to add values to specific components
- Import data from JSON file - If you have exported before some valid project data into a JSON file you can import it back by using this menu item. The big advantage of using this feature is the fact that the GRPC server API for the PCB investigator doesn't need to be connected
- Export data into JSON file - By acting this item a new file selection dialog appears to give you the possibility of choosing a target destination for exporting all loaded data into one JSON file
- Exit - Close the application (But just the graphical tool, the GRPC server itself will be still running in background as it is a separate tool)

3.2 Settings

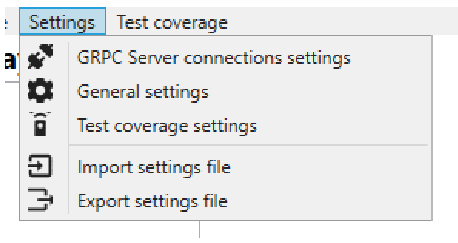

Figure 3: The settings children menu items

Within the settings menu item you will find the follwing children items:

- GRPC Server connection settings - Within this settings page you are able to define the IP address of the GRPC server API for the PCB investigator to connect to and test the connection

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- General settings - Within this settings page you are able to define the general identifier and colors of all general components from the ODB++ project
- Test coverage settings - Within this page you can define all test related identifier
- Import settings file - The possibility of importing a previously exported settings file of this application
- Export settings file - The opportunity of exporting all settings into a local file for later usage

3.3 Test coverage

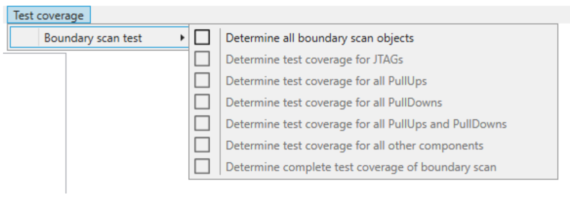

Figure 4: The test coverage children menu items

Within this menu item you actually will just find the **Boundary scan test** menu item where you can see the following children items:

- Determine all boundary scan objects - Determines in dependency on the settings from the page **Test coverage settings** all boundary scan related objects. This needs to be performed before all other operations are available
- Determine test coverage for JTAGs - This determines the test coverage for all interconnections directly between JTAG devices
- Determine test coverage for all PullUps - This determines the test coverage for all interconnections between JTAG devices and all pull up resistors
- Determine test coverage for all PullDonws - This determines the test coverage for all interconnections between JTAG devices and all pull down resistors
- Determine test coverage for all PullUps and PullDonws - This determines the test coverage for all interconnections between JTAG devices, all pull up resistors and all pull down resistors
- Determine test coverage for all other components - This determines the test coverage between all JTAG devices and all other components which are neither pull up nor pull down resistors
- Determine complete test coverage of boundary scan - This determines all previously explained tests in sum (JTAG, pull ups, pull downs and others)

4 Working with the application

Now a complete sequence of working with the application will be explained

4.1 Setting up the GRPC server connection

First of all you should either manually start the GRPC server for API with the PCB investigator or at least know its IP address of the computer where its running at. After wards you open the **Settings -> GRPC Server connections settings**

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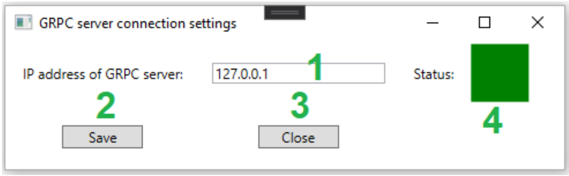

Figure 5: The GRPC server connections settings

Within this setting page you have the following possibilities:

1. IP address of GRPC server - Here you should type in the IP address of the computer where the server is running at
2. Save - By pressing this button the IP address will be saved and a connection try is being performed. If connection was successful then the painting area **3** will be painted in green otherwise in red
3. Close - Close this settings window without saving the changes made
4. Status - The connection status to the server (Green if successful otherwise red)

4.2 Setting up the general settings

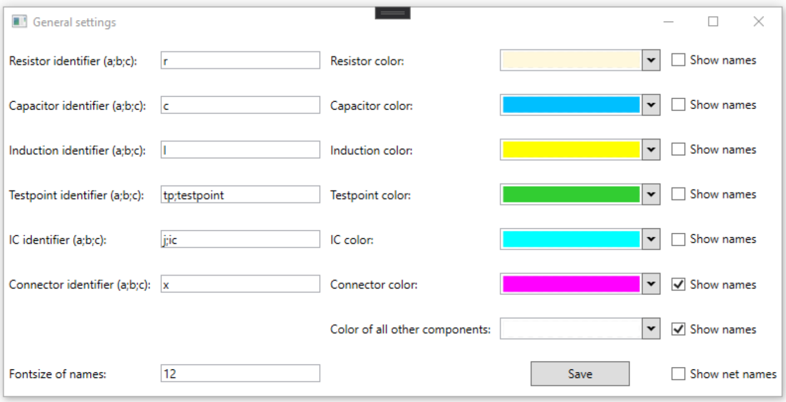

Figure 6: The general settings

By opening next the **Settings** -> **General settings** you will have the possibility of defining the following values:

- Resistor identifier (a;b;c) - The semicolon separated identifier for detecting resistors by its name
- Resistor color - The color to be used for painting resistors
- Capacitor identifier (a;b;c) - The semicolon separated identifier for detecting capacitors by its name
- Capacitor color - The color to be used for painting capacitors
- Induction identifier (a;b;c) - The semicolon separated identifier for detecting inductions by its name
- Induction color - The color to be used for painting inductions

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- Testpoint identifier (a;b;c) - The semicolon separated identifier for detecting testpoints by its name
- Testpoint color - The color to be used for painting testpoints
- IC identifier (a;b;c) - The semicolon separated identifier for detecting ICs by its name
- IC color - The color to be used for painting ICs
- Connector identifier (a;b;c) - The semicolon separated identifier for detecting connectors by its name
- Connector color - The color to be used for painting connectors
- Color of all other components - The color to be used for all other unidentified components
- Fontsize of names - The font size to be used for displaying the names of components
- Show all net names - If enabled then all net names will be displayed between active connections of components
- Save - Save all changes and close the window

4.3 Opening a ODB++ project

By clicking on **File** -> **Open ODB project folder** a new folder selection dialog will appear where you have the possibility of choosing any valid ODB++ project. After you selected one and confirmed the selection the following loading animation will be displayed for the time all data is being transferred by the GRPC server and parsed into that application:

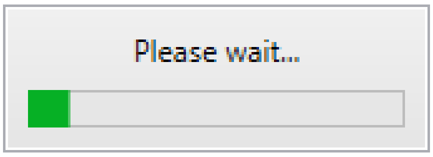

Figure 7: The loading animation

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4.4 Import BOM data file

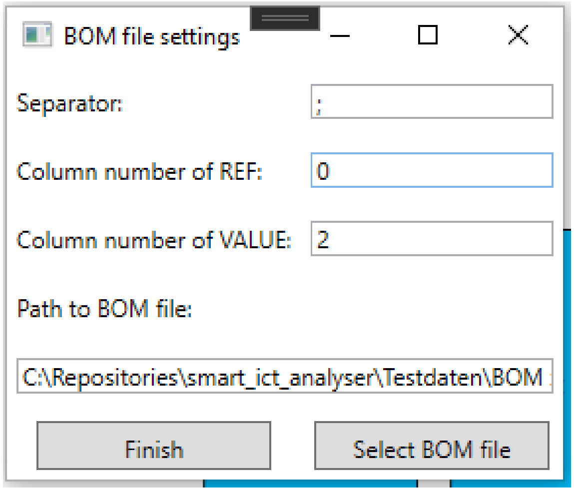

Figure 8: Import BOM data file

By clicking on the menu item **File -> Import BOM data file** a new window appears and allows you to define the following values:

- Separator - As the BOM files are typically CSV files you have to define the separator here is being used within that file
- Column number of REF - The column number (beginning with 0) where the REF, the component name is specified
- Column number of VALUE - The column number (beginning with 0) where the according value to a components name is specified
- Path to BOM file - The actual selected filepath to the file to be imported as BOM file
- Select BOM file - By clicking on this button a new file open dialog will appear to give you the possibility of choosing any file which its location then will be displayed in the field above
- Finish - Trigger the importing process of BOM data

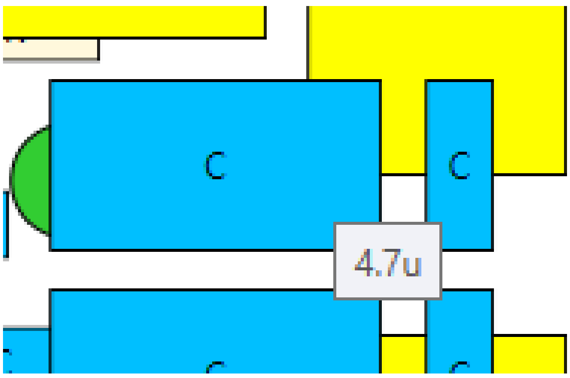

Figure 9: Values of components by hovering over them with mouse

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After successfully importing of a BOM file you should now be able to hover with the mouse over any component to see its value which was imported before. All components which does not have any values specified within the BOM file will be bordered in red color:

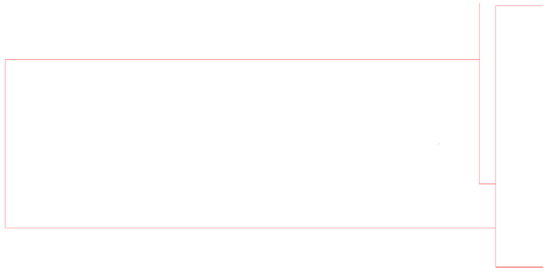

Figure 10: Red bordered components without any values from BOM file

4.5 Selection of layers for representation

After a project was loaded successfully you should then see at least one entry within the layers section of the application. By selecting one or more layers the respective objects of these selected layers will be painted into the painting area:

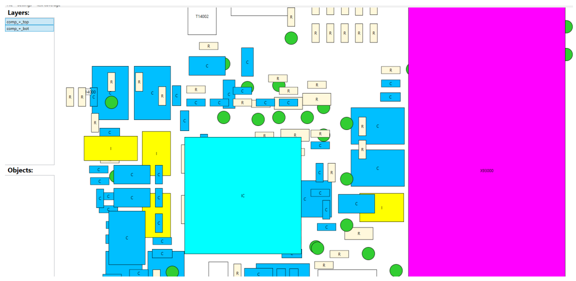

Figure 11: The area filled with selected layer objects

Within this painting area you have now these possibilites of actions:

- Mouse scrolling - By scrolling up the mouse wheel the zoom will be increased and by scrolling down the mouse wheel the zoom will be decreased
- Right click on any object of representation
  - Toggle visibility - Shows any object in transparent or not transparent way
  - Toggle visibility of all objects with the same type - Shows all objects of the same type as selected in transparent way or not
  - Toggle display of connections - Shows all connections to other objects as lines or not

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4.5.1 Toggle visibility

Figure 12: Toggle visibility of one object

As shown only the selected object will be painted in transparent way if **Toggle visibility** was performed. In that way you are able to see all objects which were covered by that specific object before

4.5.2 Toggle visibility of all objects with the same type

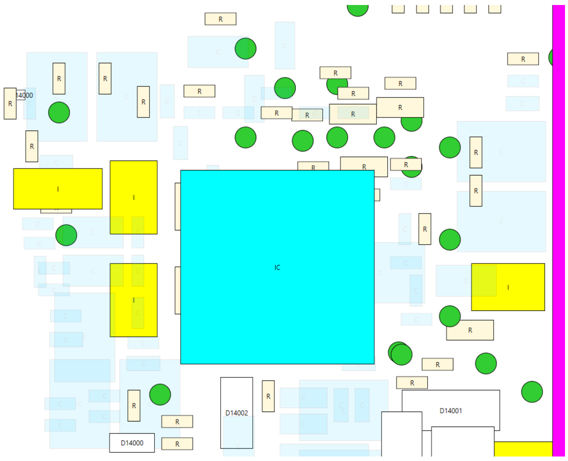

Figure 13: Toggle visibility of all objects with the same type

The same result as by performing the action **Toggle visibility** but just for every object of the same type

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4.5.3 Toggle display of connections

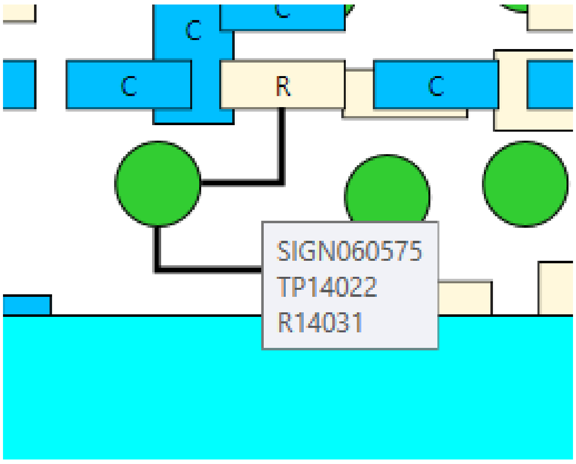

Figure 14: Toggle display of connections

By performing **Toggle display of connections** all availabe connections of that selected object will be displayed as lines. Furthermore you have the possibility of hovering over any line to get these information:

- The netname of interconnections
- The name of the first object (The selected one)
- The name of the second object

4.5.4 Show net names

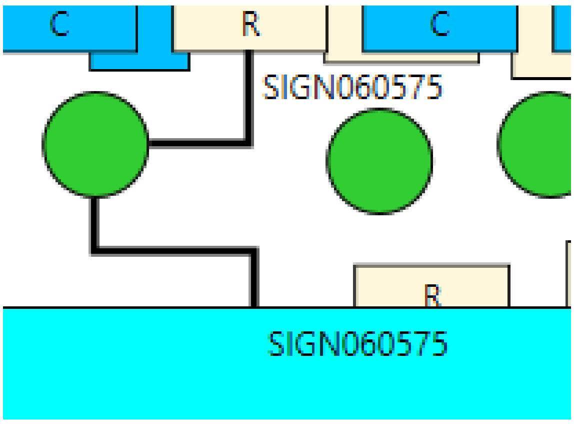

Figure 15: Show net names

If the option **Show net names** will be enabled then after toggling the visible connections furthermore all net names will be displayed

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4.6 Determine all boundary scan objects

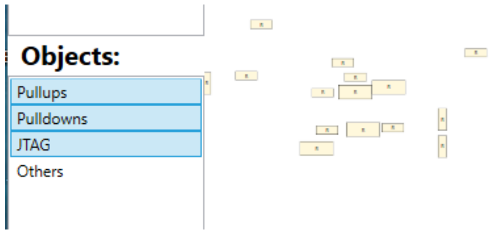

Figure 16: Object types for boundary scan test

After you performed the **Test coverage** -> **Boundary scan test** -> **Determine all boundary scan objects** action the following types will be selectable for being displayed within the painting area:

- Pullups - All pull up resistors of the project
- Pulldowns - All pull down resistors of the project
- JTAG - All JTAG related devices of the project
- Others - All other objects which are connected to the JTAG devices

4.6.1 Perform any test for determining the test coverage

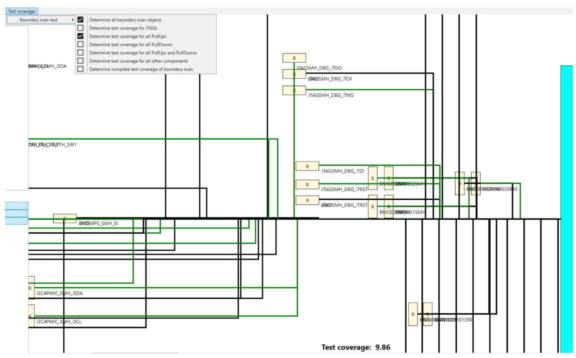

Figure 17: Test coverage of all pull ups

After you performed any test (in this example it was the test coverage for all pull ups) then the according check box to the menu item will be checked. That means if it was before unchecked then it will be checked and the other way around. Now for all components which check boxes are checked all connections of the JTAG devices will be painted in green. And finally a textual representation of the test coverage result is being displayed in the bottom area.

4.7 Export data into JSON file

After or during work with the project you shouldn't forget to export all data into a JSON file to have the possibility of loading it much faster and without the dependency on setting up the GRPC server connection.

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5 Acroyms

PullUps - Pull up resistors which are directly connected to a power supply PullDowns - Pull down resistors which are directly connected to ground connection JTAG - Joint Test Action Group (A connection device with the possibility of advanced electrical testing mechanisms) IC - Integrated circuit (A complex component consisting of many connections and other internal components)

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6 Appendix

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Revision History

Revision	Date	Author(s)	Description
1.0.0	10.02.2021	RAM	Initial version of documentation

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